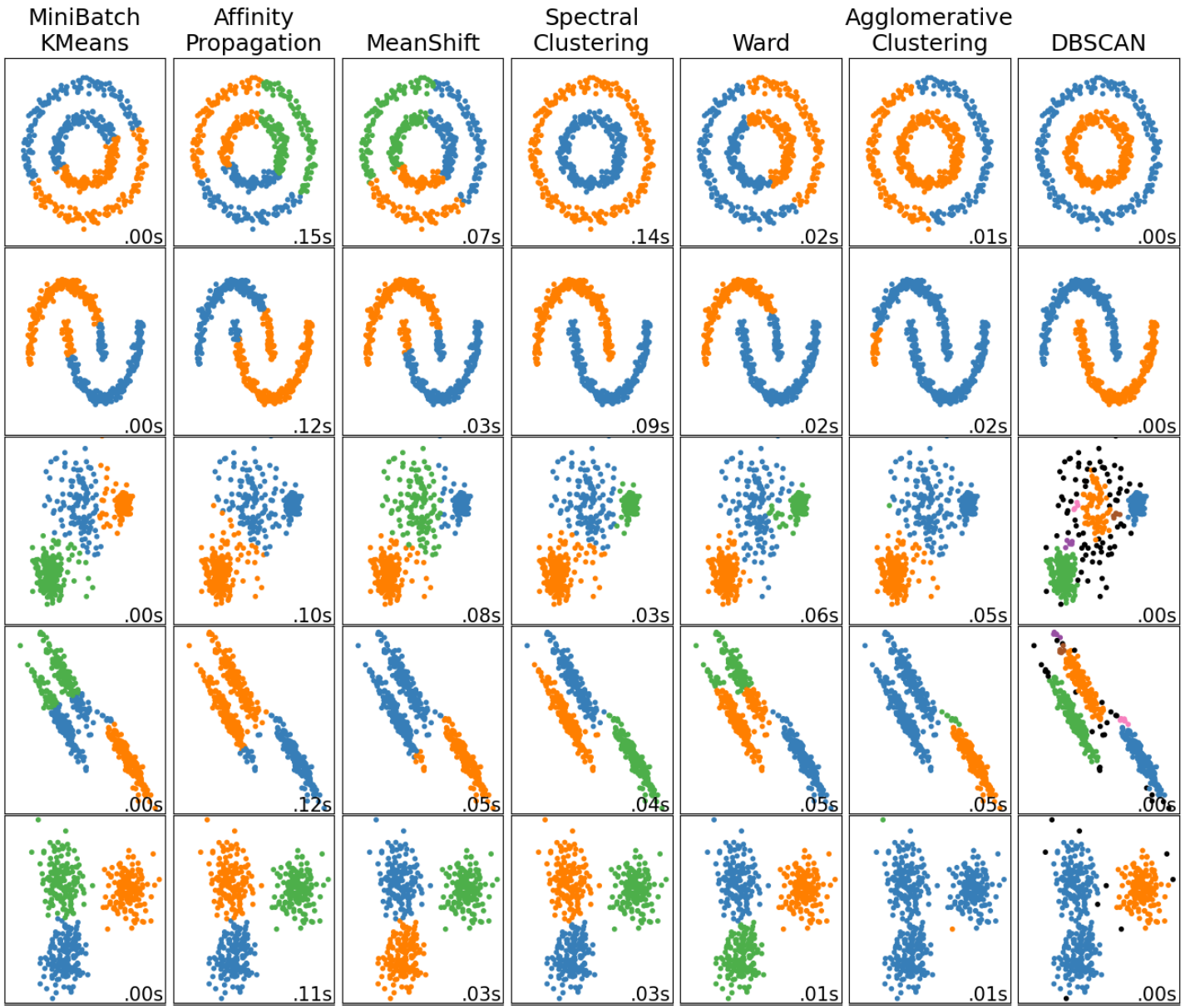
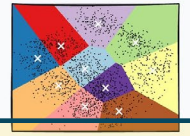
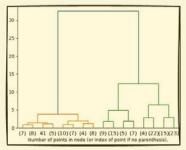
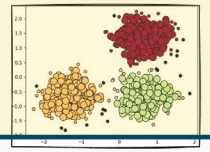
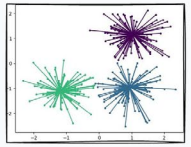
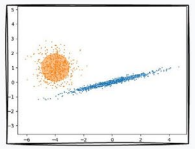
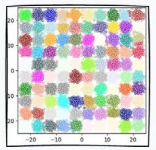
UNSUPERVISED LEARNING PART 2: HIERARCHICAL CLUSTERING



OUTLINE

- Hierarchical Clustering
  - Agglomerative
  - Divisive
- Density Based Clustering
  - DBSCAN

Clustering Algorithm Type		Clustering Methodology	Algorithm(s)
	Centroid-based	Cluster points based on proximity to centroid	KMeans KMeans++ KMedoids
	Connectivity-based	Cluster points based on proximity between clusters	Hierarchical Clustering (Agglomerative and Divisive)
	Density-based	Cluster points based on their density instead of proximity	DBSCAN OPTICS HDBSCAN
	Graph-based	Cluster points based on graph distance	Affinitiy Propagation Spectral Clustering
	Distribution-based	Cluster points based on their likelihood of belonging to the same distribution.	Gaussian Mixture Models (GMMs)
	Compression-based	Transform data to a lower dimensional space and then perform clustering	BIRCH

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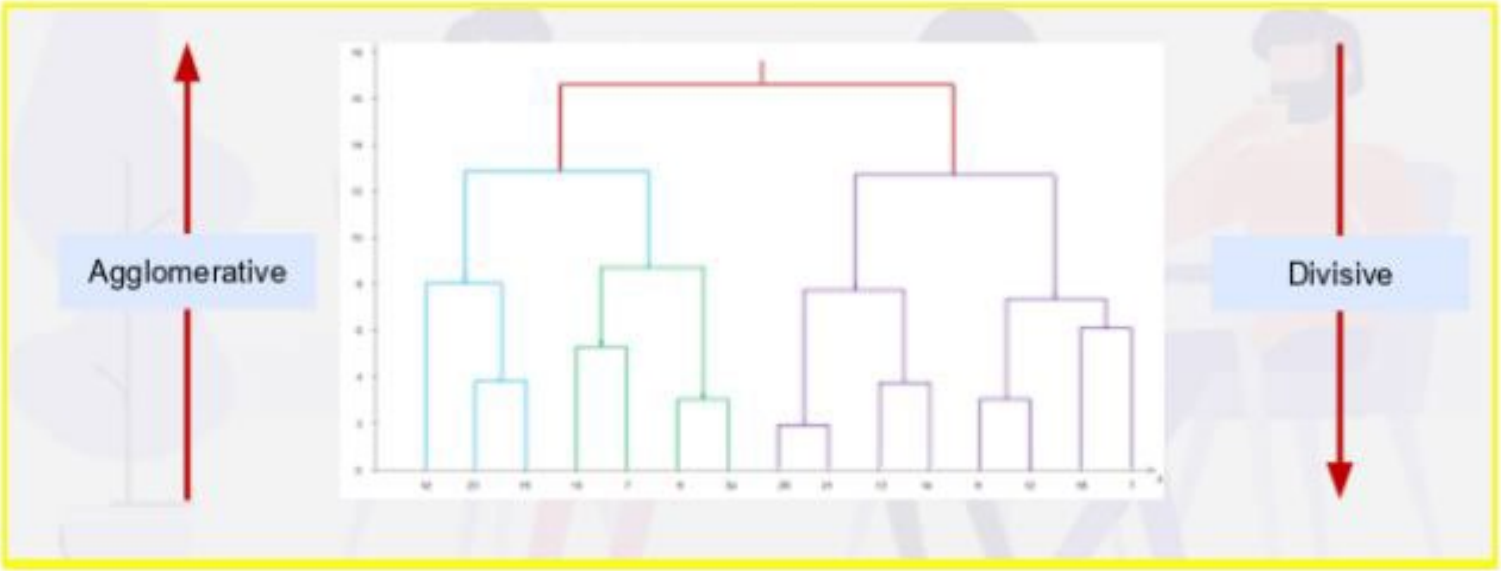
HIERARCHICAL CLUSTERING

There are two types of Hierarchical clustering techniques:

- **Agglomerative**
- **Divisive**

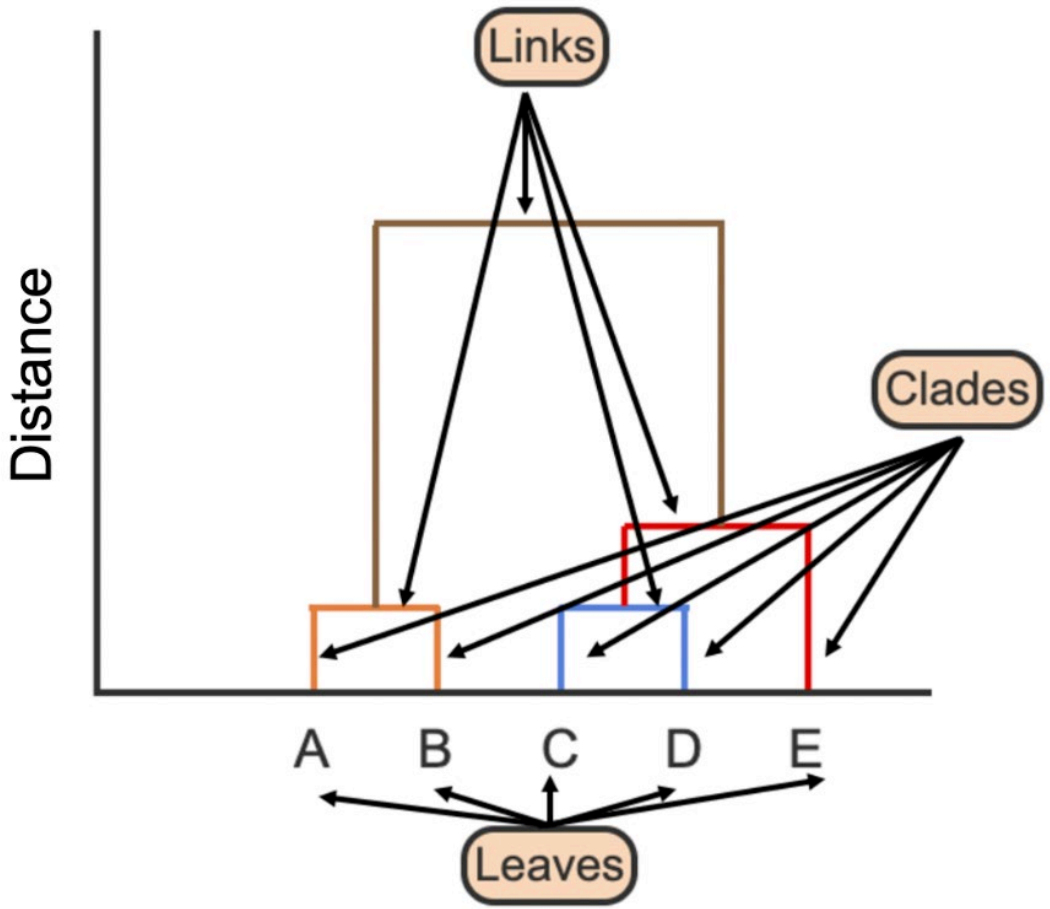
- Agglomerative clustering is a **bottom-up** approach. Start with every point as a cluster and combine points till only a single cluster. (most common)

- Divisive clustering is a **top-down** approach. Start with one large cluster and divide it into two, three, four, or more clusters.



DENDROGRAM

- The output of a hierarchical clustering algorithm is a *dendrogram*. A ***dendrogram*** is a tree that shows the order in which clusters are grouped together and the distances between clusters.
- Parts of a dendrogram are listed below:
  - A ***clade*** is a branch of a dendrogram or a vertical line.
  - A ***link*** is a horizontal line that connects two clades, whose height gives the distance between clusters.
  - A ***leaf*** is the terminal end of each clade in a dendrogram, which represents a single instance.



AGGLOMERATIVE CLUSTERING ALGORITHM

Steps:

1. Compute the proximity (distance) matrix between each point or cluster
2. Initialize: Let each data point be a cluster
3. **Repeat**
4. Merge the two closest clusters
5. Update the proximity matrix
6. **Until** only a single cluster remains

The key operation is the computation of the proximity (distance) of two clusters.

To do this, we define different approaches to measuring inter-cluster distance. These approaches are called "**Linkages**."

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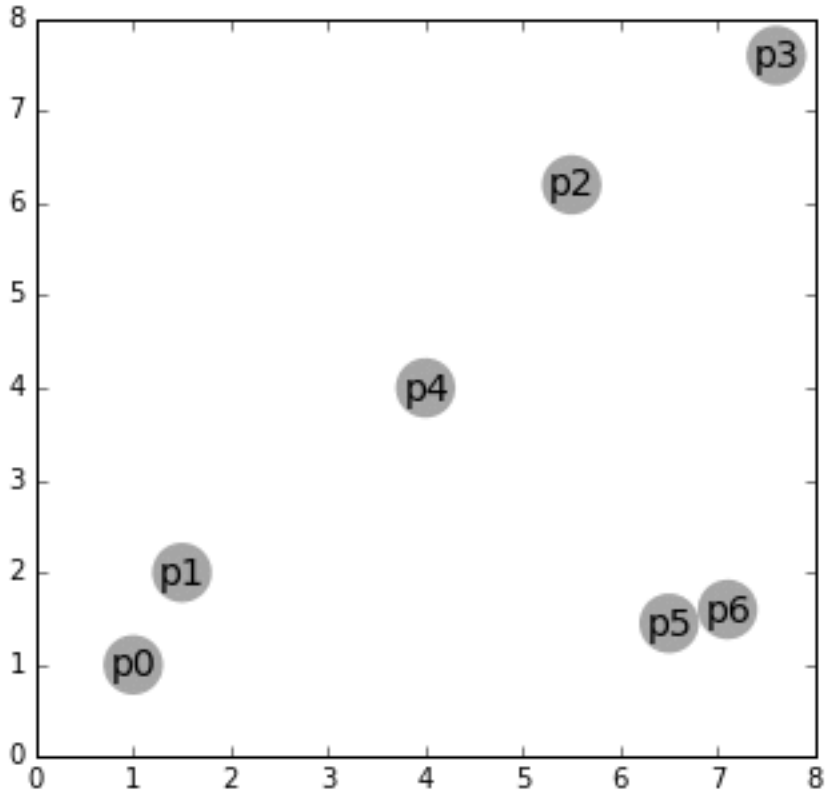
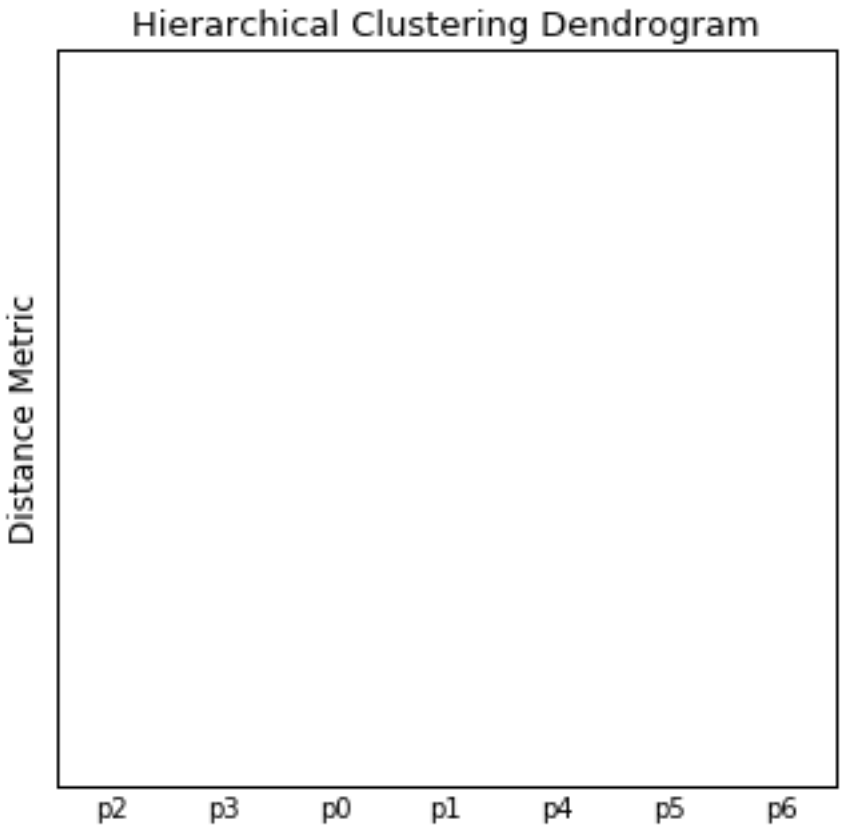
LINKAGE FUNCTION INTUITION





Hierarchical clustering is heavily influenced by two factors...

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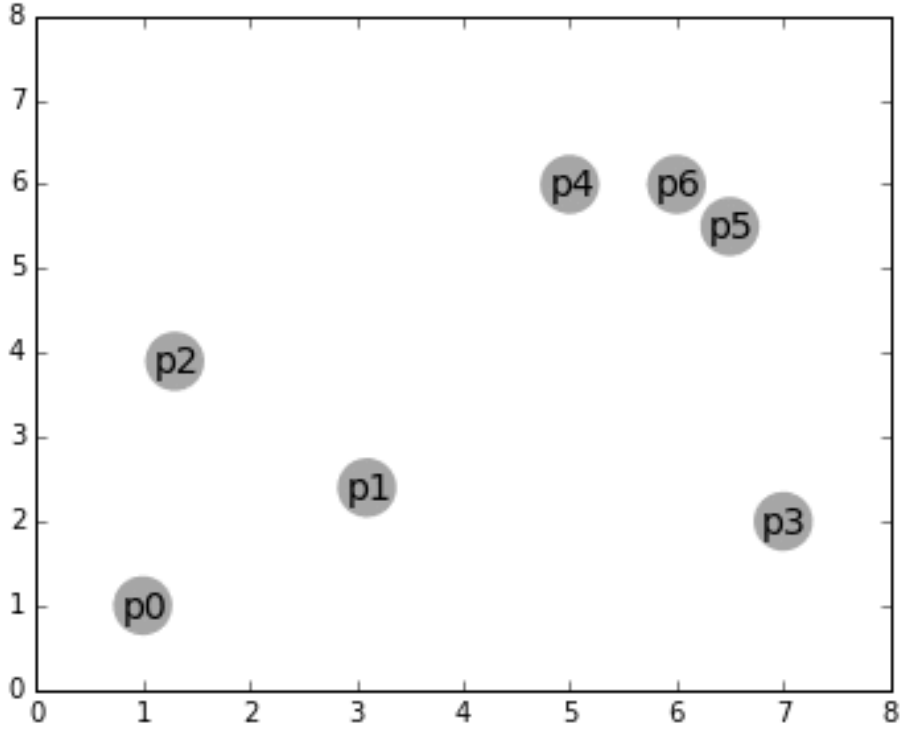
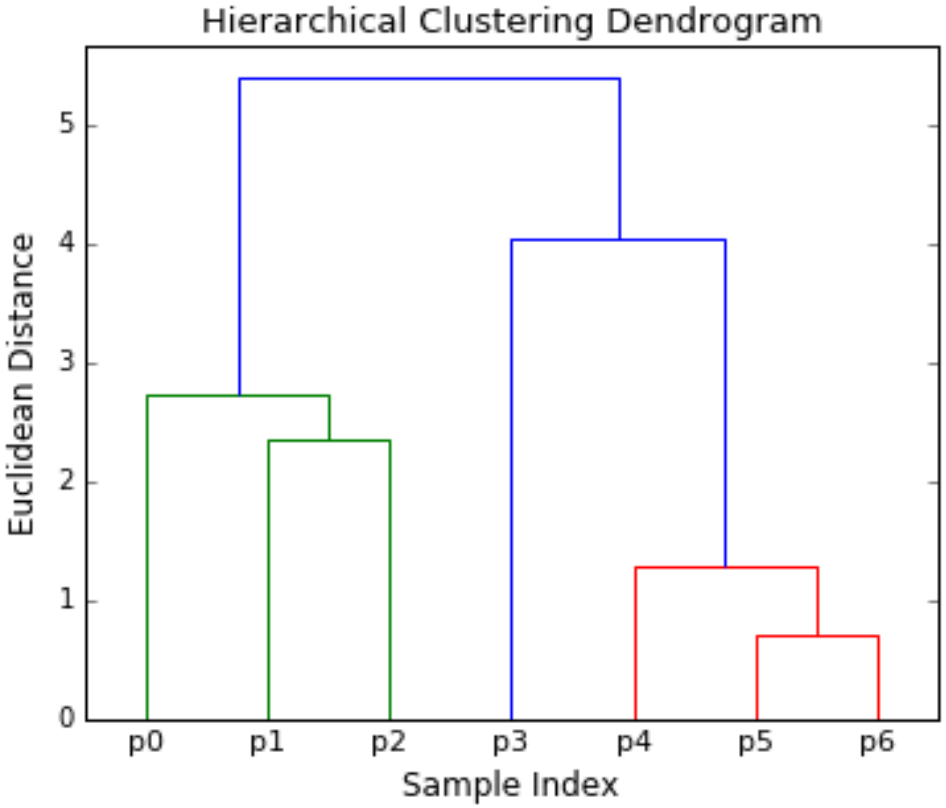
AGGLOMERATIVE CLUSTERING INTUITION

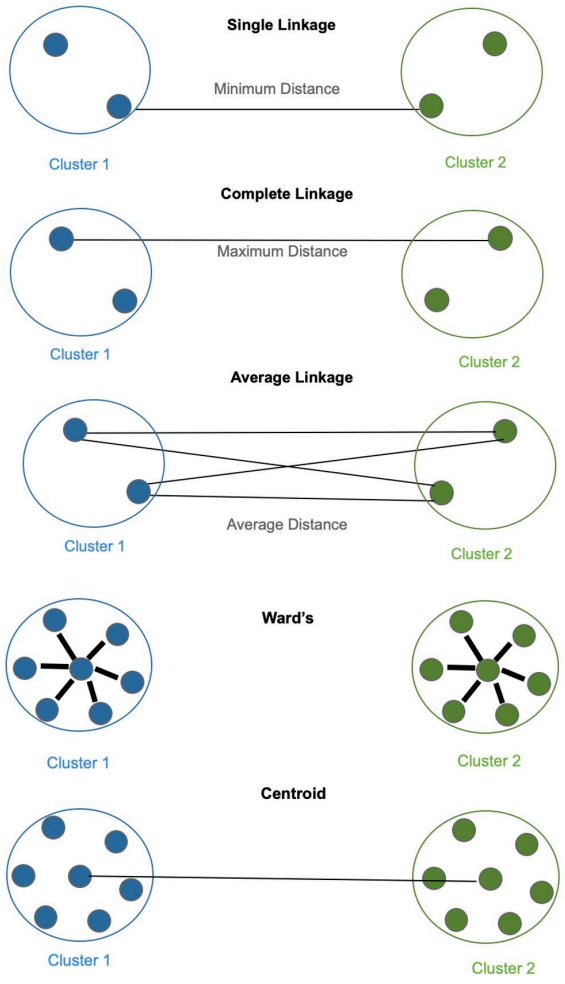




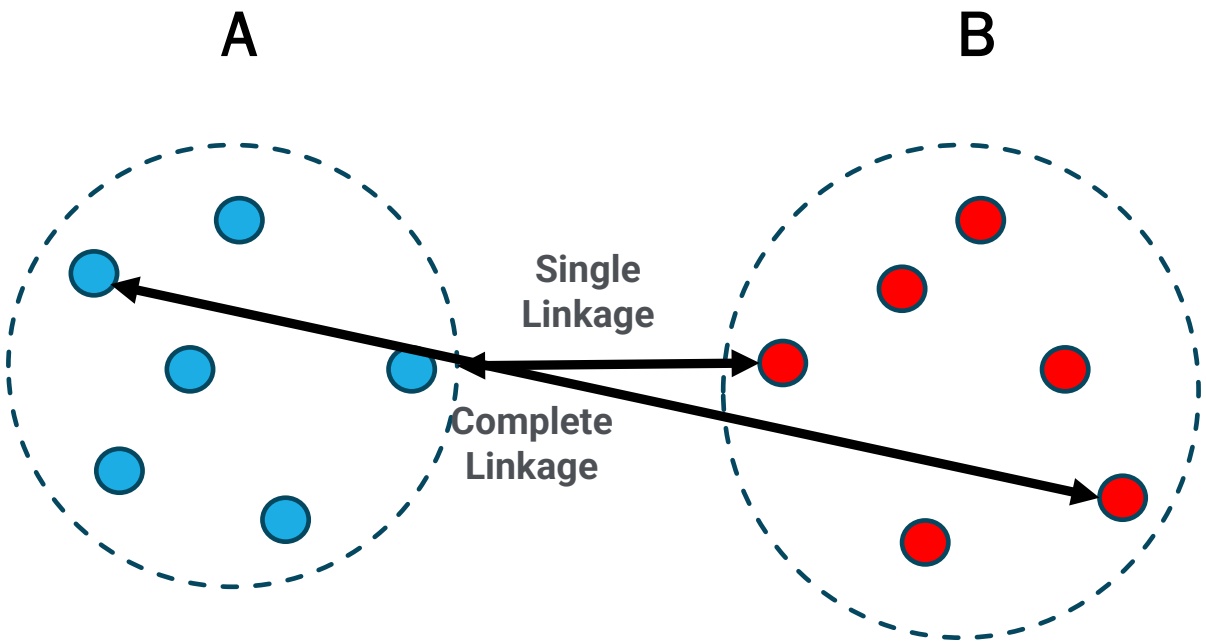
LINKAGE METHODS

Linkage methods are used to determine the distance between clusters.

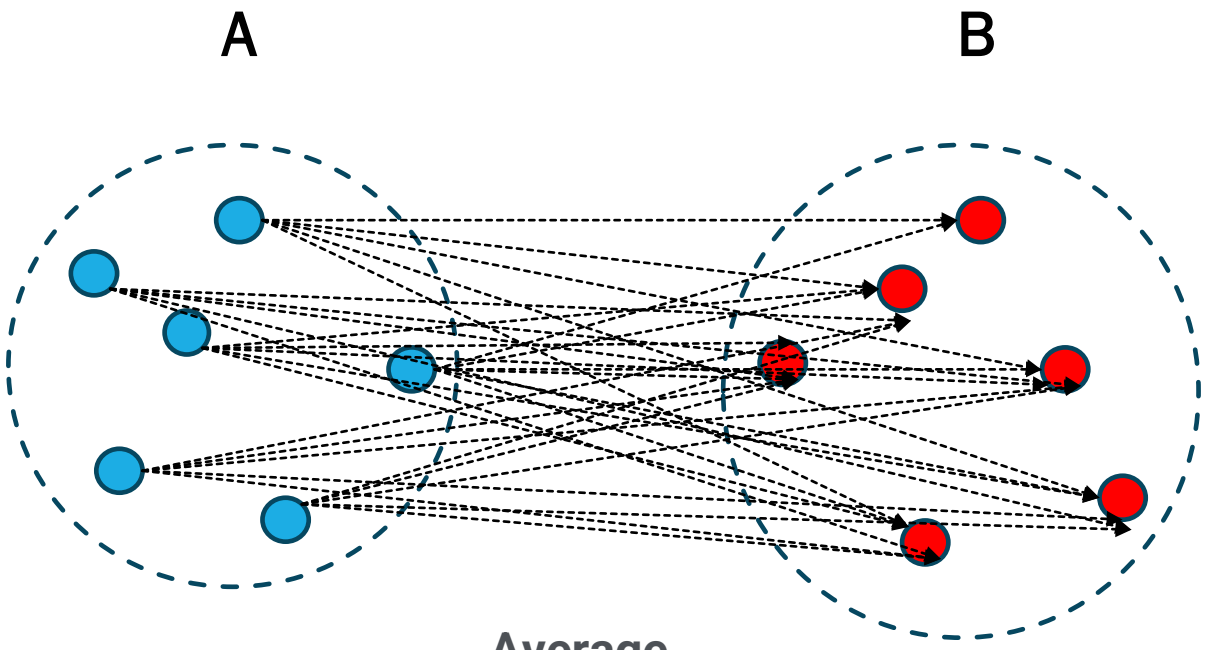
1. **Single Linkage:** (Nearest Neighbor)
2. **Complete Linkage:** Maximum distance
3. **Average Linkage:** Average distance
4. **Ward's Linkage:** Minimizes the increase in variance within the clusters.
5. **Centroid Linkage**: Minimum distance between centroids



LINKAGE METHOD EXAMPLES

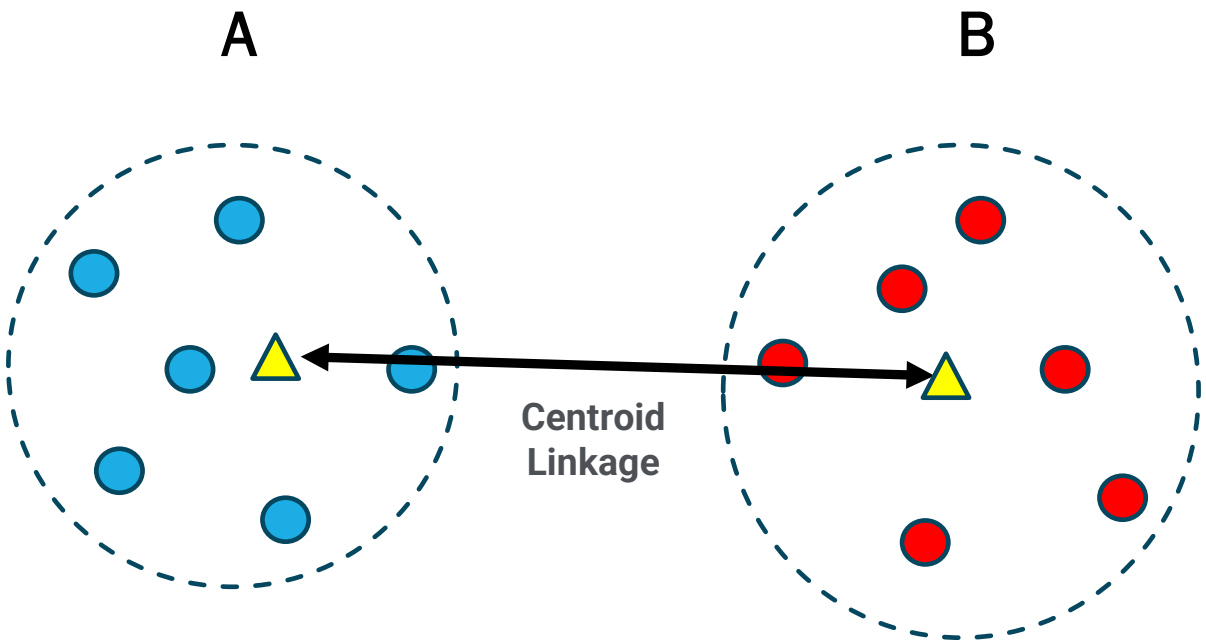


LINKAGE METHOD EXAMPLES

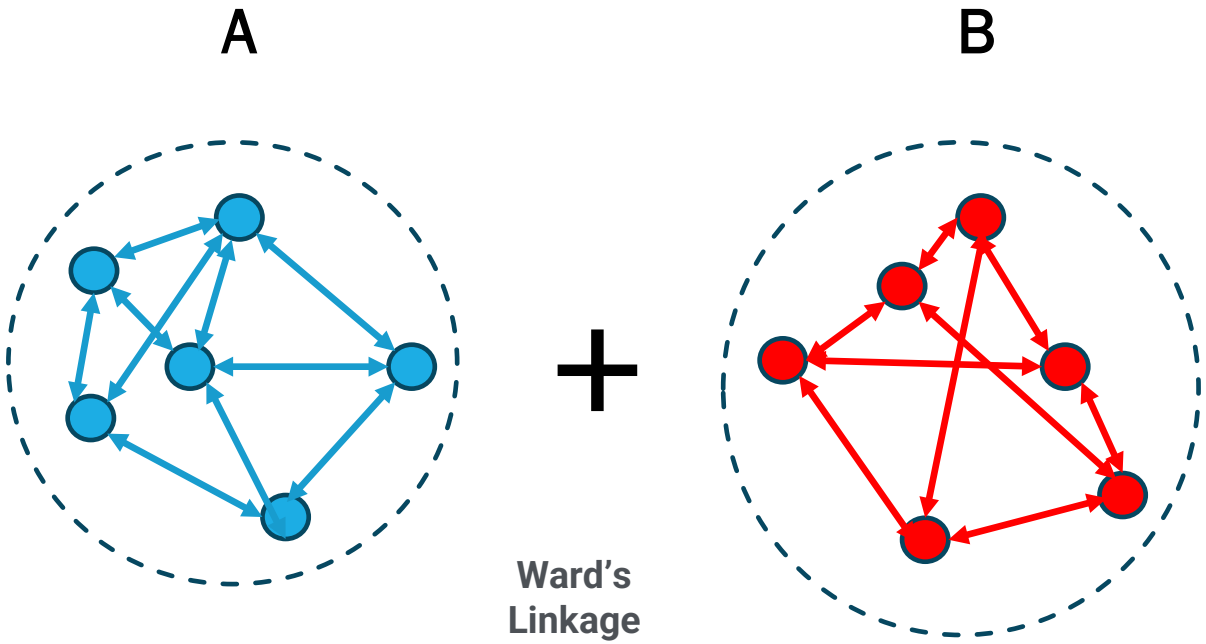


Average Linkage

LINKAGE METHOD EXAMPLES



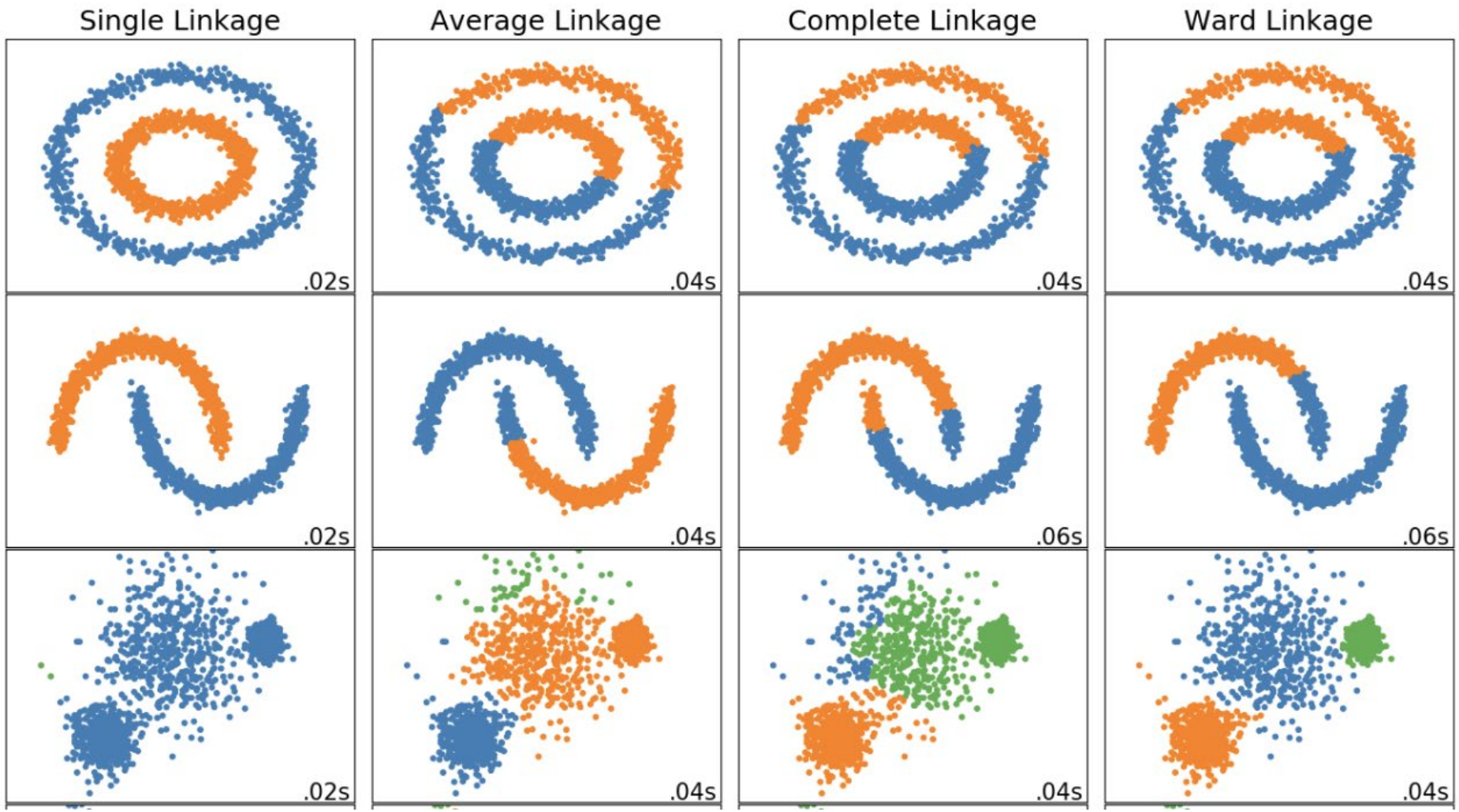
LINKAGE METHOD EXAMPLES



- We want clusters to stay tight.
- Avoid merges that create large, spread-out groups.

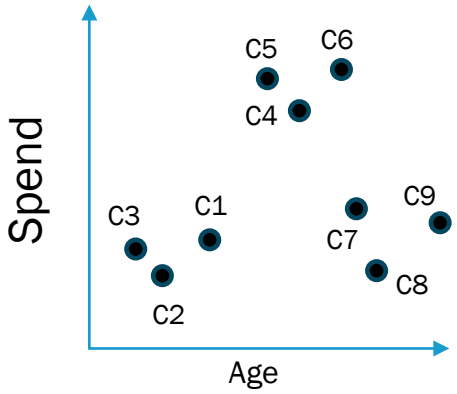
- Consider all possible pairs of clusters.
- Compute how much the total within-cluster spread would increase if we merged them.
- Merge the pair with the smallest increase.

IMPACT OF DIFFERENT LINKAGE METHODS

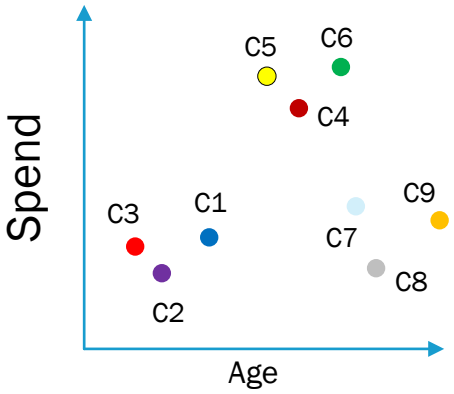


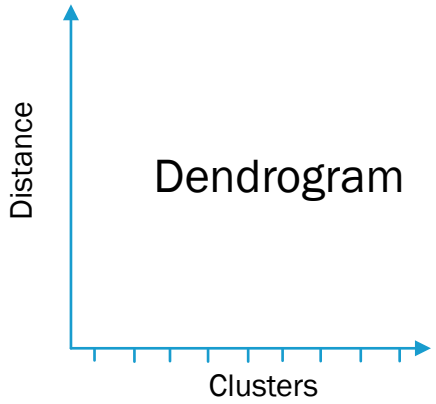
HIERARCHICAL AGGLOMERATIVE CLUSTERING EXAMPLE

Let's say a clothing store has collected the ages of 9 of its customers, labeled C1-C9, and the amount each spent at the store last month.

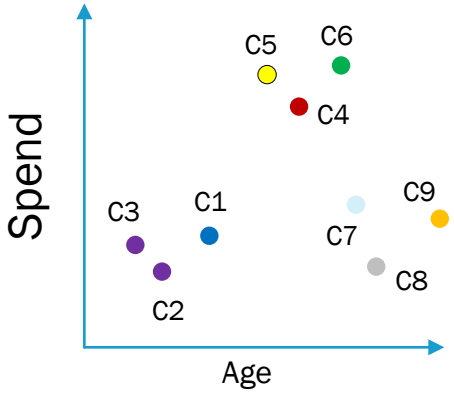


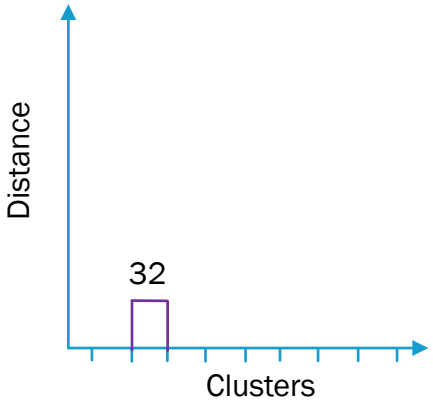
We start by making every single data point a cluster. This forms 9 clusters:



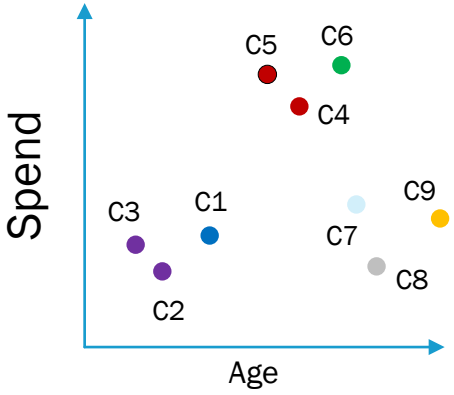


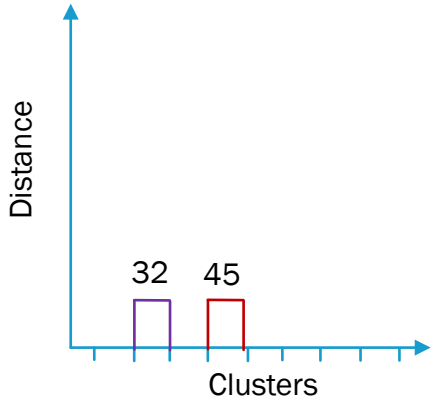
Take the two closest clusters and make them one cluster. Since C2 and C3 are closest, they form a cluster.





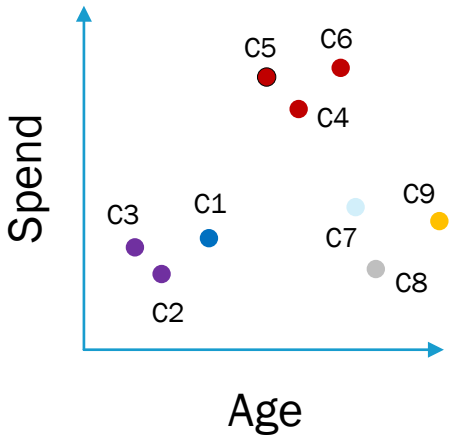
Take the next two closest clusters and make them one cluster. Since C4 and C5 are closest, they form a cluster.

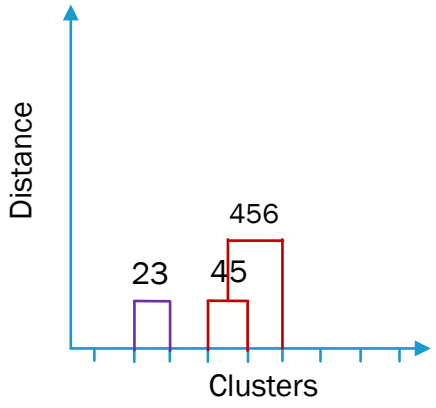




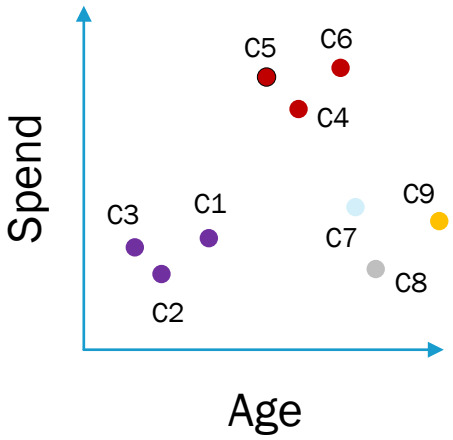
HAC EXAMPLE CONTINUED

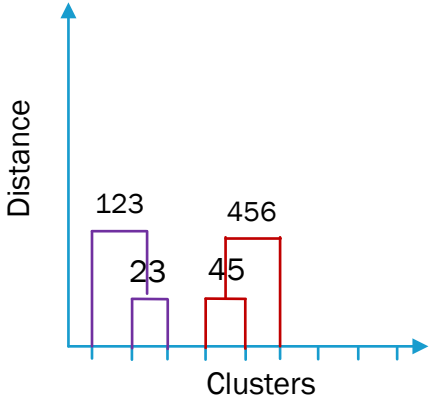
Based on Single Linkage, merge C45 with C6.



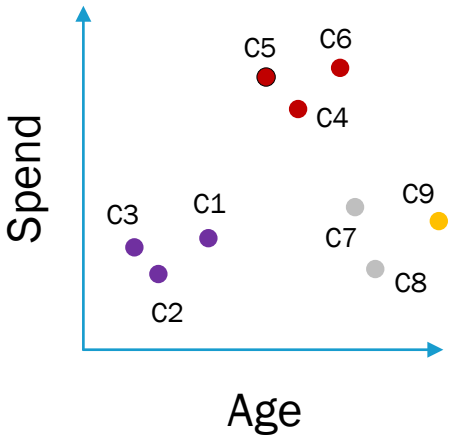


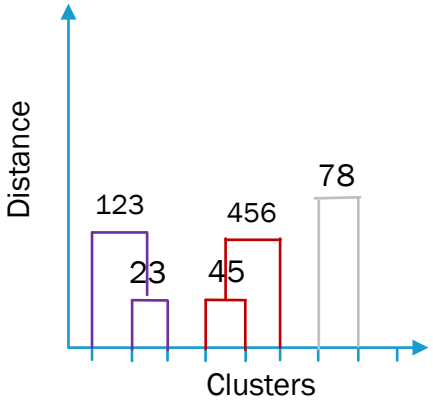
Based on Single Linkage, merge C23 with C1



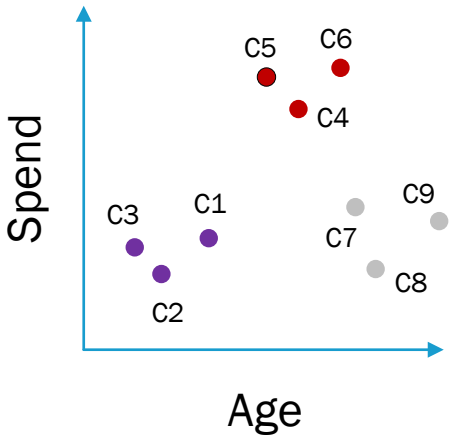


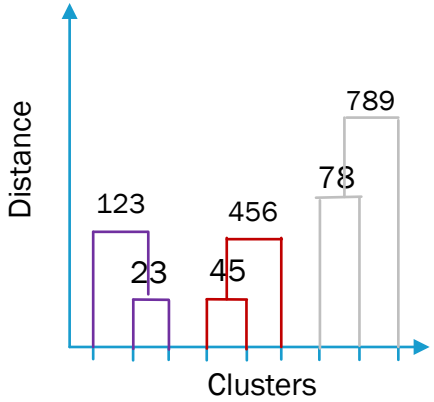
Based on distance, merge C7 and C8





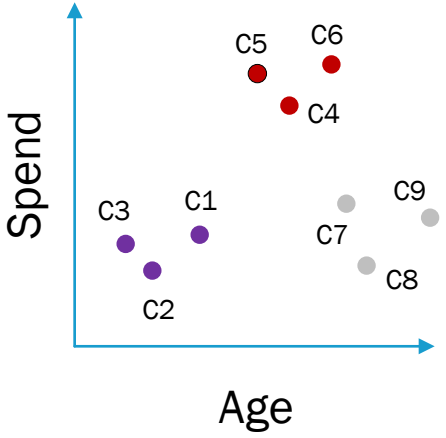
Merge C78 with C9

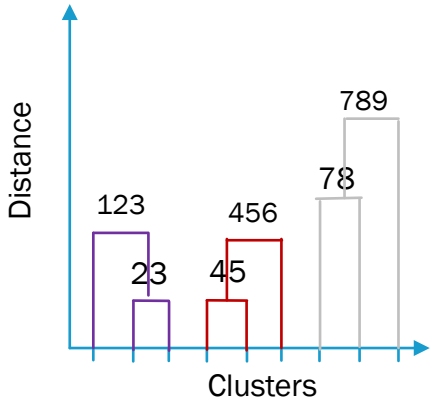




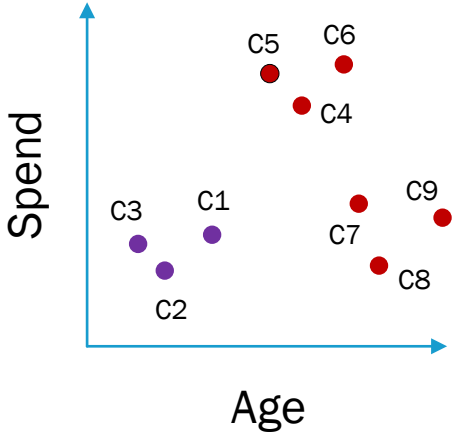
HAC EXAMPLE END

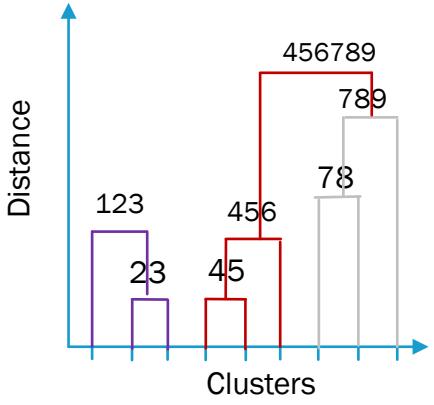
Starting from previous page



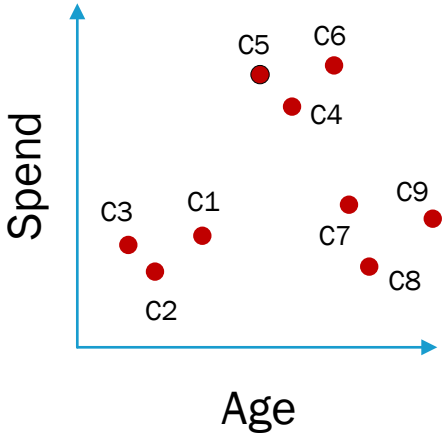


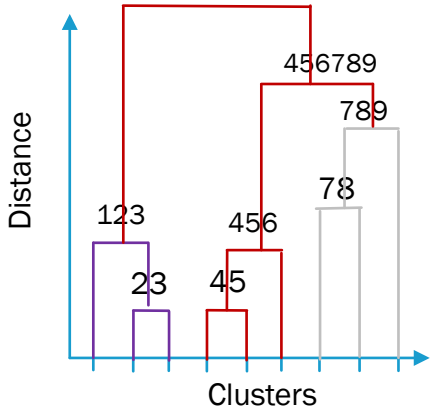
Using Single Linkage, merge C456 with C789





Finally, merge C123 with C456789 to form one cluster.

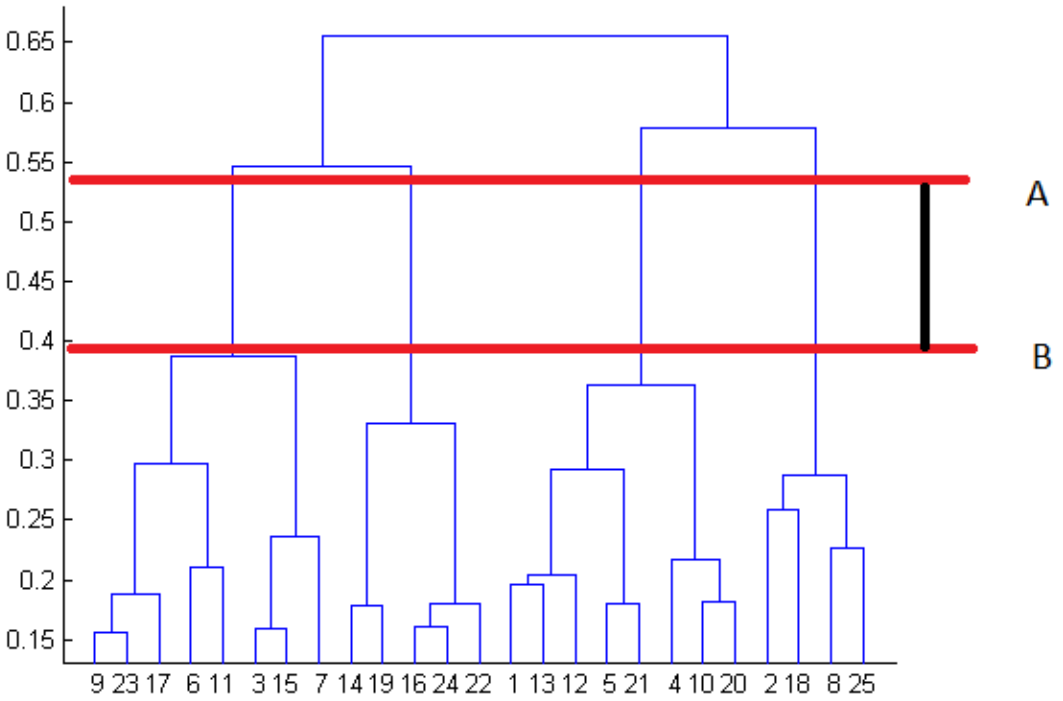




USING DENDROGRAMS TO FIND OPTIMAL CLUSTER NUMBER

We can use a dendrogram to visualize the history of groupings and figure out the optimal number of clusters.

1. Determine the largest vertical distance that doesn't intersect any of the other clusters
2. Draw a horizontal line across the diagram
3. The optimal number of clusters is equal to the number of vertical lines going through the horizontal line



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- https://github.com/nicodv/kmodes/blob/master/kmodes/kprototypes.py
- https://dashee87.github.io/data%20science/general/Clustering-with-Scikit-with-GIFs/
- https://www.geeksforgeeks.org/difference-between-agglomerative-clustering-and-divisive-clustering/
- https://stackabuse.com/dbscan-with-scikit-learn-in-python/

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